

Mock Midterm — Form B — answer key

Time allowed: 120 minutes · 100 points

QUICK KEY — MULTIPLE CHOICE

Part I: 1-D 2-D 3-D 4-B 5-C 6-B 7-A 8-C 9-C 10-D 11-A 12-B 13-C 14-A 15-A 16-D 17-C 18-B 19-C 20-A 21-B
22-D 23-D 24-B 25-A 26-B 27-C 28-C 29-B 30-C

Part I — Multiple choice

30 questions × 1 pt = 30 pts

1. MULTIPLE CHOICE

The receiver's clock runs **faster** than the sender's clock. What happens?

- A. Nothing — the receiver re-times itself automatically
- B. The receiver misses bits and sees a shorter stream
- C. The bits arrive in a different order
- D. The receiver reads extra bits (for example 1001 bps instead of 1000 bps) and misinterprets the stream ✓ correct

Why: If the receiver clock is faster or slower than the sender's, the receiver misinterprets the incoming bit stream (slide 13). With a 0.1% faster clock it receives 1001 bps instead of 1000 bps (slide 14) — extra bits, not missing ones. L03 p13; L03 p14

2. MULTIPLE CHOICE

On a constellation diagram, the horizontal (x) axis represents:

- A. time
- B. the quadrature carrier
- C. frequency
- D. the in-phase carrier ✓ correct

Why: x = in-phase (I) carrier, y = quadrature (Q) carrier. L04 p27

3. MULTIPLE CHOICE

According to the slides, what does block coding do for the line code that follows it?

- A. It lowers the bit rate by 20%
- B. It removes the need for a line code
- C. It makes every bit have a mid-bit transition
- D. It adds redundancy so that bit combinations giving DC components or poor synchronization are avoided ✓ correct

Why: Slide 48: block coding adds redundancy (extra bits), and the resulting bit stream prevents certain bit combinations that, used with line encoding, would give DC components or poor synchronization. It raises the bit rate (4B/5B: × 5/4), it does not lower it. L03 p48

4. MULTIPLE CHOICE

For a range of −10 V to +10 V and L = 4 zones, what is the zone height Δ ?

- A. 2.5 V
- B. 5 V ✓ correct
- C. 10 V
- D. 20 V

Why: $\Delta = (\max - \min)/L = (10 - (-10))/4 = 5 \text{ V}$. L03 p72

5. MULTIPLE CHOICE

A complex bandpass signal has a bandwidth of 300 kHz. What is its minimum sampling rate?

- A. 600,000 samples/s
- B. 300,000 samples/s
- C. It cannot be determined from the bandwidth alone ✓ correct
- D. 150,000 samples/s

Why: Nyquist needs the highest frequency; knowing only the width of the band does not tell us $f_{\max}$ (slide Example 8). L03 p71

6. MULTIPLE CHOICE

What does **error detection** mean in the context of line coding?

- A. The receiver corrects the corrupted bit automatically
- B. A signal pattern that is not part of the code arrives, so the receiver knows a symbol error occurred ✓ correct
- C. The signal is immune to noise and cannot be corrupted
- D. The sender repeats every bit three times

Why: Slide 15: some codes are built so that a particular signal transition is not part of the code; when it occurs, the receiver knows a symbol error occurred. Slide 16 calls a signal that cannot be corrupted “immune” to noise — stronger than error detection. L03 p15; L03 p16

7. MULTIPLE CHOICE

A band from 600 kHz to 700 kHz is shared for full-duplex ASK. Where are the two carriers?

- A. 625 kHz and 675 kHz ✓ correct
- B. 600 kHz and 700 kHz
- C. 650 kHz for both
- D. 612.5 kHz and 687.5 kHz

Why: Split the band into 600–650 and 650–700 kHz; each carrier sits in the middle of its half: 625 and 675 kHz. L04 p13

8. MULTIPLE CHOICE

A 16-QAM signal carries 20 Mbps with $d = 0$. What bandwidth does it need?

- A. 20 MHz
- B. 10 MHz
- C. 5 MHz ✓ correct
- D. 80 MHz

Why: 16-QAM: $r = \log_2 16 = 4$ bits per element $\rightarrow S = 20/4 = 5$ Mbaud $\rightarrow B = (1 + 0) \times 5 \text{ MHz} = 5 \text{ MHz}$. L04 p26

9. MULTIPLE CHOICE

What is the propagation time over a 3,000 km link if signals travel at 2×10^8 m/s?

- A. 1.5 ms
- B. 6 ms
- C. 15 ms ✓ correct
- D. 150 ms

Why: Propagation time = distance / propagation speed = $(3,000 \times 1,000 \text{ m}) / (2 \times 10^8 \text{ m/s}) = 0.015 \text{ s} = 15 \text{ ms}$. L02 p35; L02 p37

10. MULTIPLE CHOICE

According to Shannon, the capacity of a channel depends on:

- A. the bandwidth and the number of signal levels
- B. the number of signal levels and the SNR
- C. the propagation speed and the distance
- D. the bandwidth and the signal-to-noise ratio ✓ correct

Why: $C = B \times \log_2(1 + \text{SNR})$. There is no signal-level term: the levels of the signal do not dictate the capacity of the channel. L02 p31

11. MULTIPLE CHOICE

A noisy channel has a bandwidth of 2 MHz and an SNR of 15. What is its Shannon capacity?

- A. 8 Mbps ✓ correct
- B. 30 Mbps
- C. 4 Mbps
- D. 2 Mbps

Why: $C = B \log_2(1 + \text{SNR}) = 2 \text{ MHz} \times \log_2 16 = 2 \text{ MHz} \times 4 = 8 \text{ Mbps}$. L02 p31; L02 p33

12. MULTIPLE CHOICE

In **MLT-3**, the signal is at +V and the next bit is a 1. Where does the level go?

- A. -V
- B. 0 V ✓ correct
- C. It stays at +V
- D. +2V

Why: MLT-3: a 0 keeps the level; a 1 moves $\pm V \rightarrow 0$, and $0 \rightarrow$ the opposite of the last nonzero level. From +V a 1 goes to 0 V (the next 1 would then go to -V), so 1111 cycles + 0 - 0 (slide 44). L03 p44

13. MULTIPLE CHOICE

A digital signal has 16 levels. How many bits does each level carry?

- A. 16 bits
- B. 8 bits
- C. 4 bits ✓ correct
- D. 2 bits

Why: A signal with L levels needs $\log_2 L$ bits for each level: $\log_2 16 = 4$ bits. L02 p14

14. MULTIPLE CHOICE

The two styles of media access control named on the slides are:

- A. ALOHA style and token style ✓ correct
- B. TCP and UDP
- C. Analog and digital
- D. Simplex and duplex

Why: Slide 13: ALOHA style and token style. L01 p13

15. MULTIPLE CHOICE

Which pattern does **B8ZS** substitute for eight consecutive zeros?

- A. 000VB0VB ✓ correct
- B. B00V
- C. 000V
- D. VB0VB000

Why: B8ZS uses 000VB0VB, where V is a violation of the line-encoding rule and B follows the bipolar rule (slide 59). B00V and 000V are the HDB3 patterns for four zeros. L03 p59

16. MULTIPLE CHOICE

Which component of a PCM decoder smooths the held staircase back into an analog signal?

- A. A comparator
- B. A delay unit
- C. A quantizer
- D. A low-pass filter ✓ correct

Why: The hold circuit makes a staircase; a low-pass filter with cutoff = $f_{\max}$ smooths it into the analog signal. L03 p81

17. MULTIPLE CHOICE

Why is a **DC component** a problem for line coding?

- A. It makes the baud rate higher than the bit rate
- B. It needs an extra wire to carry the clock
- C. Most channels are bandpass and may not support the low frequencies that a constant level creates ✓ correct
- D. It makes every bit have a mid-bit transition

Why: When the voltage stays constant for long periods, the low-frequency content of the signal increases. Most channels are bandpass and may not pass low frequencies, so the DC component has to be removed (slide 12). L03 p12

18. MULTIPLE CHOICE

Increasing the number of quantization levels L ...

- A. raises the quantization error
- B. lowers the quantization error but raises the bit rate ✓ correct
- C. lowers both the error and the bit rate
- D. has no effect on the bit rate

Why: More zones → smaller Δ → smaller error, but more bits per sample → higher bit rate. L03 p77

19. MULTIPLE CHOICE

In the 4B/5B table, which 5-bit word replaces the data word **0000**?

- A. 01001
- B. 11101
- C. 11110 ✓ correct
- D. 10101

Why: From the table on slide 51: 0000 → 11110. For comparison 0001 → 01001, 0011 → 10101 and 1111 → 11101; slide 52 shows 0000, 0001 and 1111 being substituted. L03 p51; L03 p52

20. MULTIPLE CHOICE

A signal of 6 MHz travels in free space ($c = 3 \times 10^8$ m/s). What is its wavelength?

- A. 50 m ✓ correct
- B. 5 m
- C. 0.02 m
- D. 18 m

Why: $\lambda = c / f = (3 \times 10^8 \text{ m/s}) / (6 \times 10^6 \text{ Hz}) = 50 \text{ m}$. L02 p9; L02 p10

21. MULTIPLE CHOICE

During a phone call you faintly hear a conversation from a neighbouring wire. This is:

- A. thermal noise
- B. crosstalk ✓ correct
- C. impulse noise
- D. induced noise

Why: Crosstalk is signals jumping from one wire to another, normally between adjacent wires. L02 p24

22. MULTIPLE CHOICE

To carry 5 bits per signal element, how many distinct signal elements are needed?

- A. 5
- B. 10
- C. 25
- D. 32 ✓ correct

Why: $L = 2^r = 2^5 = 32$. L04 p7

23. MULTIPLE CHOICE

Why can a composite signal be **distorted** on its way through a medium?

- A. The noise power is larger than the signal power
- B. An amplifier adds the same gain to every frequency
- C. It loses some of its energy in overcoming the resistance of the medium
- D. Its frequency components travel at different propagation speeds and arrive at different times ✓ correct

Why: Distortion changes the signal's form or shape because the components of a composite signal arrive at different times. Loss of energy is attenuation; unwanted signals are noise. L02 p23

24. MULTIPLE CHOICE

Which of these is a WAN technology mentioned on the slides?

- A. Bluetooth
- B. Submarine fiber cable ✓ correct
- C. A hub
- D. A coaxial patch cord

Why: WANs use long-distance fiber such as submarine cable, plus satellite and terrestrial microwave. L01 p10

25. MULTIPLE CHOICE

A data stream of rate N is sent with **2B1Q**. What is the average signal (baud) rate?

- A. $N/4$ ✓ correct
- B. $N/2$
- C. N
- D. $2N$

Why: Each signal element carries 2 bits ($r = 2$) and the average case has $c = 1/2$, so $S = c \times N \times 1/r = 1/2 \times N \times 1/2 = N/4$ (slides 38 and 46). The $r = 1/2$ printed in the slide's box is a mistake. L03 p38; L03 p46

26. MULTIPLE CHOICE

A link has a bandwidth of 10 Mbps. What can be said about the throughput through it?

- A. It is always equal to 10 Mbps
- B. It is always less than 10 Mbps ✓ correct
- C. It can be greater than 10 Mbps
- D. It is unrelated to the bandwidth

Why: Slide 34: a link may have a bandwidth of B bps, but we can only send T bps through it, with T always less than B . L02 p34

27. MULTIPLE CHOICE

In the slides' example of digital transmission, a binary 1 can be represented as:

- A. -5 V
- B. 0 V
- C. $+5$ V ✓ correct
- D. a 5 kHz tone

Why: Slide 15: a binary 1 can be $+5$ V and a binary 0 can be 0 V. L01 p15

28. MULTIPLE CHOICE

Which of these is NOT a service the data link layer provides to the upper layers?

- A. Framing
- B. Flow control
- C. Routing packets between networks ✓ correct
- D. Error detection

Why: Routing packets between networks is the network layer's job; the data link layer works between adjacent nodes. L01 p5

29. MULTIPLE CHOICE

SpaceX Starlink satellites are deployed at about what altitude?

- A. 55 km
- B. 550 km ✓ correct
- C. 5,500 km
- D. 35,786 km

Why: Slide 11: a global deployment of satellites at 550 km above the earth. L01 p11

30. MULTIPLE CHOICE

In binary FSK, bit 1 is sent at which frequency?

- A. $f_c - \Delta f$
- B. f_c
- C. $f_c + \Delta f$ ✓ correct
- D. $2f_c$

Why: Slide 14: "1" $\rightarrow f_1 = f_c + \Delta f$; "0" $\rightarrow f_2 = f_c - \Delta f$. L04 p14

Part II — Identification

20 questions × 1 pt = 20 pts

1. IDENTIFICATION

The device that converts digital signals into analog signals for transmission (its partner converts them back).

Answer: Modulator (also accepted: modem, modulator/demodulator)

Why: A modulator converts digital to analog; a demodulator converts analog back to digital. L01 p17

2. IDENTIFICATION

The difference in delay (latency) from one packet to another.

Answer: Jitter (also accepted: delay jitter)

Why: Jitter is the difference in delay or latency per packet. L02 p34

3. IDENTIFICATION

The polar NRZ version in which the *change or lack of change* in polarity gives the bit: a 1 inverts the level, a 0 does not.

Answer: NRZ-I (also accepted: NRZ-Inversion, NRZ inversion, Non-Return-to-Zero Inversion, Non Return to Zero Inversion, NRZ-Invert, NRZ inverted, Non-Return-to-Zero Inverted)

Why: NRZ-I (NRZ-Inversion): the inversion or lack of inversion determines the bit. In NRZ-L the level itself determines the bit (slide 24). L03 p22

4. IDENTIFICATION

The data link layer service that ensures sufficient bandwidth is available and that delays are predictable and guaranteed.

Answer: Quality of service (also accepted: QoS, quality of services)

Why: Selecting QoS parameters for a transmission. L01 p5

5. IDENTIFICATION

The main frequency of a composite signal (in the slide example the other sine waves have 3 and 9 times this frequency).

Answer: Fundamental frequency (also accepted: fundamental)

Why: The main frequency of a composite signal is normally called the fundamental frequency (f ; the others are $3f$ and $9f$). L02 p12

6. IDENTIFICATION

The data link layer service that keeps frames in the correct order as they are exchanged.

Answer: Frame sequencing (also accepted: sequencing)

Why: Frame sequencing maintains the correct ordering of frames. L01 p5

7. IDENTIFICATION

In the B8ZS pattern 000VB0VB, what does the letter **V** stand for?

Answer: Violation (also accepted: a violation, violation pulse, violate)

Why: V stands for violation — a pulse that violates the line-encoding rule (it repeats the polarity of the previous pulse); B stands for bipolar — a pulse that follows the rule (slide 59). L03 p59

8. IDENTIFICATION

The type of network that uses unguided media and can be short or long distance.

Answer: Wireless network (also accepted: wireless)

Why: Slide 6: wireless networks — unguided media connection, short or long distance. L01 p6

9. IDENTIFICATION

Serial transmission in which uneven gaps between frames are not allowed; bits flow with fixed, equal gaps.

Answer: Isochronous transmission (also accepted: isochronous)

Why: Isochronous transmission cannot have uneven gaps between frames. L03 p96

10. IDENTIFICATION

Sending a digital signal over a channel without changing it into an analog signal.

Answer: Baseband transmission (also accepted: baseband)

Why: Baseband transmission needs a wide-bandwidth medium, a dedicated medium whose bandwidth constitutes only one channel. L02 p16

11. IDENTIFICATION

The digital-to-analog scheme that varies only the phase of the carrier and is much more robust than ASK.

Answer: PSK (also accepted: phase shift keying)

Why: PSK keeps amplitude and frequency constant; noise mostly affects amplitude. L04 p21

12. IDENTIFICATION

The property of a line code that lets the receiver keep its clock in step with the sender's clock by using the transitions in the signal itself.

Answer: Self-synchronization (also accepted: self synchronizing, self-synchronizing, self-clocking, self clocking, selfsynchronization)

Why: Clocks at the sender and the receiver must have the same bit interval (slide 13); a self-synchronizing code carries the timing in its transitions. Scrambling also aims at "self clocking" (slide 57). L03 p13

13. IDENTIFICATION

The drift of the receiver's running average (the baseline) caused by a long string of 0s or 1s, which makes correct decoding difficult.

Answer: Baseline wandering (also accepted: baseline wander)

Why: The baseline is the receiver's running average of the received signal power; long runs of one level make it drift (slide 11). L03 p11

14. IDENTIFICATION

The unit engineers use to show that a signal has lost or gained strength.

Answer: Decibel (also accepted: dB, deciBel, db)

Why: The decibel (dB) measures the relative strengths of two signals, or of one signal at two points: negative if attenuated, positive if amplified. L02 p20

15. IDENTIFICATION

The device in an FSK transmitter whose output frequency follows its input voltage.

Answer: Voltage-controlled oscillator (also accepted: VCO, voltage controlled oscillator)

Why: A VCO turns the NRZ input into a frequency change, which is binary FSK. L04 p18

16. IDENTIFICATION

The name of the sampled signal — pulses whose heights are the (non-integer) sample values.

Answer: PAM (also accepted: pulse amplitude modulation, PAM signal)

Why: Sampling is called pulse amplitude modulation; its output still has analog values. L03 p64

17. IDENTIFICATION

The PCM decoder circuit that holds each pulse amplitude until the next pulse arrives.

Answer: Hold circuit (also accepted: hold, sample and hold circuit)

Why: The hold circuit turns the pulses into a staircase, which the low-pass filter then smooths. L03 p81

18. IDENTIFICATION

The French mathematician who showed that a composite signal is a combination of simple sine waves with different frequencies, amplitudes and phases.

Answer: Fourier (also accepted: Jean-Baptiste Fourier, Joseph Fourier, Jean-Baptiste Joseph Fourier)

Why: Fourier showed that any composite signal is a sum of simple sine waves. (The slide says "early 1900s"; his work dates from the early 1800s.) L02 p11

19. IDENTIFICATION

The line-coding family in which *all* signal levels are on one side of the time axis (NRZ is the example).

Answer: Unipolar (also accepted: unipolar NRZ, unipolar encoding, unipolar coding, unipolar scheme)

Why: Slide 20: unipolar schemes keep every level above (or below) the axis; NRZ is the example. It is simple but costly in power, and prone to baseline wandering and DC components. L03 p20

20. IDENTIFICATION

The kind of FSK that switches frequency without adhering to the current phase of the signal.

Answer: Non-coherent FSK (also accepted: noncoherent FSK, non coherent FSK, non-coherent, noncoherent)

Why: Non-coherent FSK can jump in phase at the switch; coherent FSK switches only at the same phase. L04 p17

Part III — Draw: line encodings and waves

4 problems × 5 pts = 20 pts

1. PROBLEM · 5 PTS · L03A

Draw the bit stream **11010010** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

- **2B1Q**: one level per pair of bits from the transition table (previous level positive: 00 → +1, 01 → +3, 10 → -1, 11 → -3; previous level negative: the signs reversed); assume the level before the first pair is positive.
- **MLT-3**: a 0 keeps the level; a 1 moves round the cycle 0 → +V → 0 → -V → ...; start at 0 V with the last nonzero level -V, so the first 1 goes to +V.
- **RZ**: a 1 is +V then 0 V, a 0 is -V then 0 V (every bit returns to zero in the middle).

1. **2B1Q** — one level per pair of bits from the transition table (previous level positive: 00 → +1, 01 → +3, 10 → -1, 11 → -3; previous level negative: the signs reversed); assume the level before the first pair is positive. L03 p38

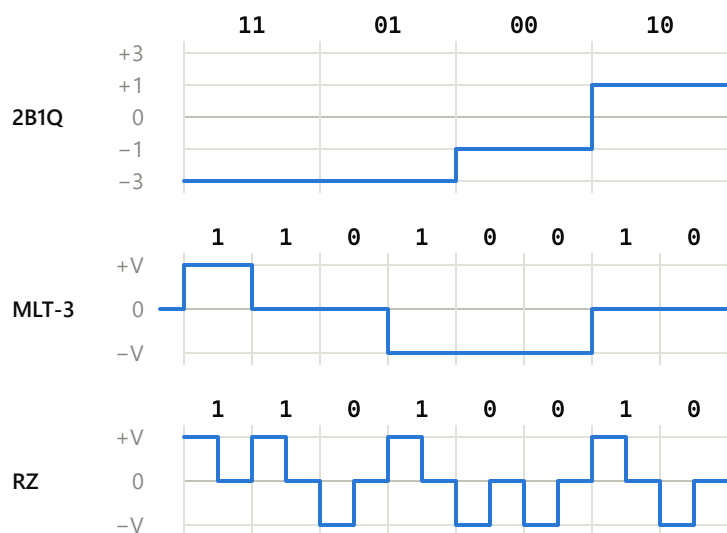
2. Bit by bit: 11→-3, 01→-3, 00→-1, 10→+1 — the waveform is -3 -3 -1 +1.

3. **MLT-3** — a 0 keeps the level; a 1 moves round the cycle 0 → +V → 0 → -V → ...; start at 0 V with the last nonzero level -V, so the first 1 goes to +V. L03 p44

4. Bit by bit: 1→+, 1→0, 0→0, 1→-, 0→-, 0→-, 1→0, 0→0 — the waveform is + 0 0 - - - 0 0.

5. **RZ** — a 1 is +V then 0 V, a 0 is -V then 0 V (every bit returns to zero in the middle). L03 p27

6. Bit by bit: 1→+0, 1→+0, 0→-0, 1→+0, 0→-0, 0→-0, 1→+0, 0→-0 — the waveform is (+0)(+0)(-0)(+0)(-0)(-0)(+0)(-0).

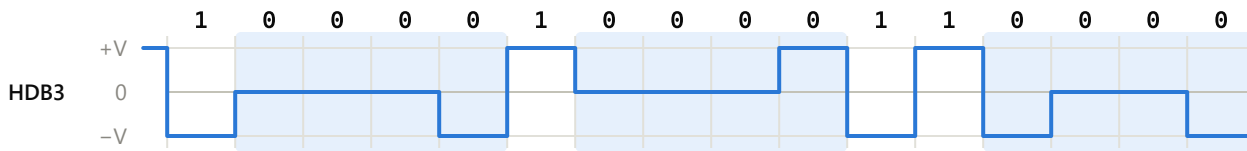


2. PROBLEM · 5 PTS · L03A

Apply **HDB3** scrambling (on top of AMI) to the data stream **1000 0100 0011 0000** and draw the line signal. The last nonzero pulse sent before this stream was positive (+V). At the start of the stream the number of nonzero pulses since the last substitution is even.

1. Scrambling works on top of AMI: 0 = 0 V and the 1s alternate in polarity. The last pulse before the stream was positive (+V), so the next 1 is -. L03 p58
2. HDB3 replaces four consecutive 0s: count the nonzero pulses since the last substitution — **even** → **B00V**, **odd** → **000V** (the total becomes even either way). L03 p60
3. Substitution 1 (bits 2–5): 1 nonzero pulse since the start → odd → **000V** = 0 0 0 - (V repeats the previous pulse). L03 p61

4. Substitution 2 (bits 7–10): 1 nonzero pulse since the last substitution → odd → **000V** = 0 0 0 + (V repeats the previous pulse). L03 p61
5. Substitution 3 (bits 13–16): 2 nonzero pulses since the last substitution → even → **B00V** = - 0 0 - (B alternates from the previous pulse, V repeats B). L03 p61
6. The transmitted pulses are - 0 0 0 - + 0 0 0 + - + - 0 0 - — no run of four zeros is left.



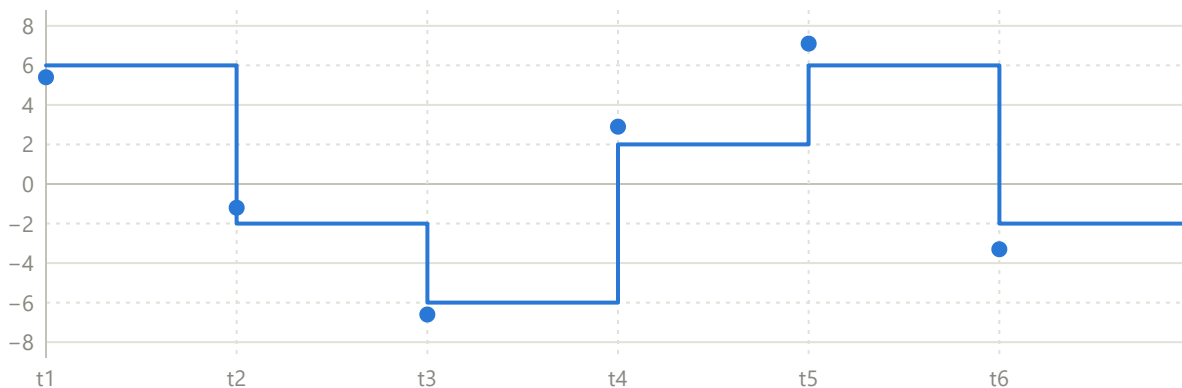
Pulse before the stream: +. Substitutions: bits 2–5 → 000V (odd pulses since last substitution); bits 7–10 → 000V (odd pulses since last substitution); bits 13–16 → B00V (even pulses since last substitution). Marks: V at bit 5, V at bit 10, B at bit 13, V at bit 16.

3. PROBLEM · 5 PTS · L03B

A signal between **-8 V** and **8 V** is quantized into **L = 4** zones (zone height $\Delta = 4$ V). Its samples at $t_1 \dots t_6$ are **5.4, -1.2, -6.6, 2.9, 7.1, -3.3 V**. Draw the quantized signal: each sample moved to its zone midpoint and held until the next sample.

1. Zone height $\Delta = (\max - \min)/L = (8 - (-8))/4 = 4$ V; the zone midpoints are -6, -2, 2, 6 V. L03 p72
2. Each sample goes to the midpoint of its zone: 5.4 → 6, -1.2 → -2, -6.6 → -6, 2.9 → 2, 7.1 → 6, -3.3 → -2. L03 p76
3. Holding each value until the next sample arrives (as the decoder's hold circuit does) turns the samples into a staircase. L03 p81

Quantized signal



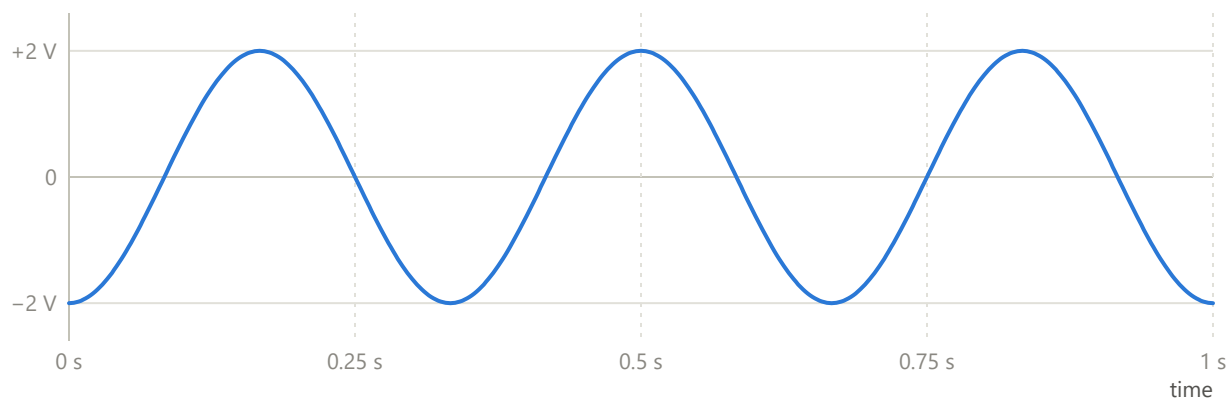
- Each sample moved to its zone midpoint: 5.4 → 6, -1.2 → -2, -6.6 → -6, 2.9 → 2, 7.1 → 6, -3.3 → -2 V.
- Flat steps: each value held for one sample interval (a staircase that changes only at the sampling instants).

4. PROBLEM · 5 PTS · L02

Draw one second of a sine wave with peak amplitude **2 V**, frequency **3 Hz** and phase **270°**.

1. Period: $T = 1/f = 1/3 \text{ Hz} = 1/3 \text{ s} (\approx 0.333 \text{ s})$, so 3 cycles fit in 1 s. L02 p7
2. Phase 270°: the wave starts **at the trough**. L02 p8
3. Peak amplitude 2 V: the crests reach +2 V and the troughs -2 V. L02 p7

Sine wave, 0 to 1 s



- At $t = 0$ the wave starts at the trough (phase 270°).
 - 3 full cycles in 1 s — one cycle every $T = 1/f = 1/3 \text{ s} (\approx 0.333 \text{ s})$.
 - Crests at +2 V and troughs at -2 V.
-

Part IV — Problem solving (no calculator)

2 problems × 5 pts = 10 pts

1. PROBLEM · 5 PTS · L02

We have a channel with a **4,000 Hz** bandwidth. The SNR for this channel is **63**. Find the upper limit of the bit rate. For better performance we then use a lower bit rate of **16,000 bps**: how many signal levels are needed?

1. The channel is noisy, so Shannon gives the upper limit: $C = B \times \log_2(1 + \text{SNR}) = 4,000 \text{ Hz} \times \log_2(1 + 63) = 4,000 \text{ Hz} \times \log_2 64 = 4,000 \text{ Hz} \times 6 = \mathbf{24,000 \text{ bps}}$. L02 p31 L02 p33
2. We may send at most C, and a lower rate is safer: $N = 16,000 \text{ bps} \leq 24,000 \text{ bps}$. L02 p33
3. Nyquist links the rate to the levels: $N = 2 \times B \times \log_2 L$, so $\log_2 L = 16,000 / (2 \times 4,000) = 2$. L02 p29
4. $L = 2^2 = \mathbf{4}$ signal levels. L02 p33

Upper limit C = 24 kbps Signal levels L = 4

2. PROBLEM · 5 PTS · L04

We have an available bandwidth of **120 kHz which spans from 500 kHz to 620 kHz**. What should be the carrier frequency and the bit rate if we modulate our data with **FSK**, $d = 0.5$, and $2\Delta f = 30 \text{ kHz}$? Also give the frequencies used for bit 1 and bit 0.

1. Middle of the band: $f_c = (500 \text{ kHz} + 620 \text{ kHz}) / 2 = \mathbf{560 \text{ kHz}}$. L04 p16
2. FSK: $B = (1 + d) \times S + 2\Delta f$. L04 p15
3. $120 \text{ kHz} = (1 + 0.5) \times S + 30 \text{ kHz} \Rightarrow S = (120 \text{ kHz} - 30 \text{ kHz}) / 1.5 = 60 \text{ kbaud}$.
4. Binary FSK has $r = 1$, so $N = S = \mathbf{60 \text{ kbps}}$.
5. Bit 1 $\rightarrow f_c + \Delta f = \mathbf{575 \text{ kHz}}$; bit 0 $\rightarrow f_c - \Delta f = \mathbf{545 \text{ kHz}}$. L04 p14

Carrier $f_c = 560 \text{ kHz}$ Bit rate $N = 60 \text{ kbps}$ f_1 (bit 1) = 575 kHz f_2 (bit 0) = 545 kHz

Part V — Essay (draw and/or explain)

4 questions × 5 pts = 20 pts

1. ESSAY

Explain the purpose of the data link layer and describe at least four services it provides to the upper layers. Then explain why digital data is sometimes sent as an analog signal.

RUBRIC

1 pt Purpose: formats data into frames and moves them error-free between two end nodes over the physical layer.

2 pts At least four services described correctly: framing, frame sequencing, flow control, error detection (sometimes correction), QoS, link provisioning.

2 pts Analog transmission: needed when bandwidth is limited — the digital signal is modulated onto an analog signal (modulator) and demodulated at the receiver.

MODEL ANSWER

The **data link layer** regulates and formats the transmission of data from a node onto the network cabling. It dictates data formats, timing and bit sequencing so that **frames** travel **error-free between two end nodes** over the physical layer.

Services: **framing** (partitioning data into frames with recognized boundaries); **frame sequencing** (keeping frames in order); **flow control** (an acceptable flow across the link); **error detection** (detecting physical-layer errors and notifying, sometimes correcting them); and **QoS** (ensuring sufficient bandwidth and predictable, guaranteed delays). It also provisions links between adjacent nodes.

Digital data is sent as an analog signal when the **bandwidth is limited**: a medium with limited bandwidth cannot pass the digital signal, so a **modulator** turns it into an analog signal for transmission and a **demodulator** recovers the digital data at the other end.

L01 pp4–5; L01 p17

2. ESSAY

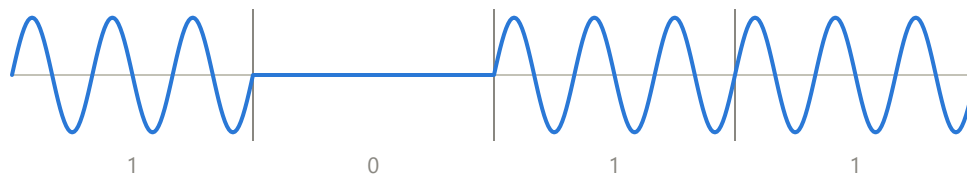
Compare ASK, FSK and PSK: what each changes in the carrier and how each one's bandwidth depends on the baud rate; then say how robust ASK and PSK are to noise. Then explain how QPSK sends twice the bits of BPSK without needing more bandwidth. Sketches of the three signals for a short bit string are welcome.

RUBRIC

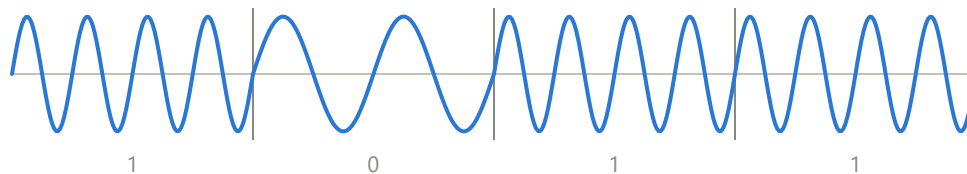
- 1 pt** States what each scheme varies: ASK amplitude, FSK frequency, PSK phase.
- 1 pt** Bandwidths: ASK/PSK $B = (1 + d)S$; FSK $B = (1 + d)S + 2\Delta f$ (extra spacing).
- 1 pt** Noise: ASK is more vulnerable (noise changes amplitude); PSK is much more robust than ASK.
- 2 pts** QPSK: 2 bits per element on I and Q carriers 90° apart $\rightarrow$ same baud rate (same bandwidth), double bit rate ($N = r \times S$).

MODEL ANSWER

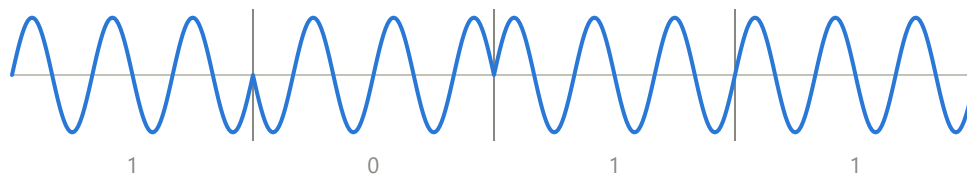
ASK (OOK)



Binary FSK



BPSK



ASK changes the carrier's amplitude (OOK: 1 = carrier, 0 = none), **FSK** its frequency (bit 1 $\rightarrow f_c + \Delta f$, bit 0 $\rightarrow f_c - \Delta f$), and **PSK** its phase (BPSK: 0° and 180°).

Bandwidth follows the baud rate S : ASK and PSK need $B = (1 + d)S$, while FSK needs extra room for the two carriers, $B = (1 + d)S + 2\Delta f$.

Noise mostly disturbs amplitude, so ASK is more vulnerable; PSK keeps a constant amplitude and is much more robust than ASK.

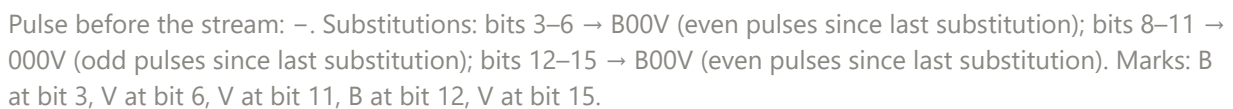
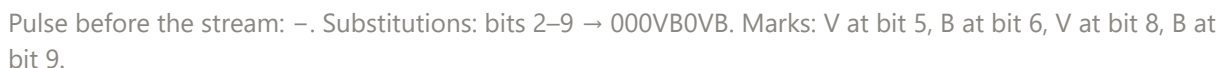
QPSK splits the bits into pairs: the first bit drives the in-phase carrier, the second the quadrature carrier (90° apart), and the sum gives one of four phases. Each element carries $r = 2$ bits, so at the same baud rate — and therefore the same bandwidth — it carries twice the bit rate of BPSK ($N = r \times S$).

Why does **AMI** need scrambling? Explain how **B8ZS** and **HDB3** work, and draw B8ZS for the bits **100000000** when the last pulse sent before this stream was $-V$ (so the 1 is sent as $+V$). Mark each V and B.

- 1 pt** Problem: a long run of 0s in AMI is a flat 0 V line, so the receiver loses synchronization; scrambling fixes it without increasing the bandwidth.
- 1 pt** B8ZS replaces eight consecutive 0s with 000VB0VB: V (violation) has the same polarity as the pulse before it, B (bipolar) follows the AMI alternation.
- 1 pt** HDB3 replaces four consecutive 0s: 000V if the number of nonzero pulses since the last substitution is odd, B00V if it is even.
- 1 pt** The receiver recognizes the violations and turns each pattern back into 0s; scrambling keeps the signal self-clocking with no low frequencies and no wider bandwidth (slide 57).
- 1 pt** Drawing: + 0 0 0 + - 0 - + (the 1 is + because the pulse before was -; then 000VB0VB with V = +, B = -, V = -, B = +).

In AMI a 0 is sent as 0 V, so a long run of 0s is a flat line with no transitions and the receiver loses synchronization. **Scrambling** replaces such runs with a pattern containing deliberate **violations** that the receiver can recognize; unlike block coding it does not increase the bandwidth. L03 p57

For 100000000 after a $-V$ pulse, the 1 is sent as $+V$; the eight 0s become 000V0VB0VB = 0 0 0 + - 0 - +.



L03 pp57-61

4. ESSAY

Describe the three transmission impairments named in the lecture: **attenuation**, **distortion** and **noise**. For each, draw (or describe) what it does to a signal and say what causes it; then explain how losses, gains and noise are measured.

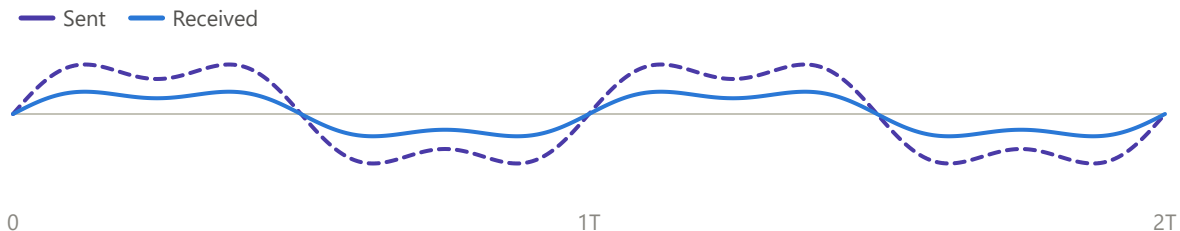
RUBRIC

- 1 pt** Attenuation: loss of energy as the signal overcomes the resistance of the medium; amplifiers compensate.
- 1 pt** Distortion: the signal changes its form or shape; the components of a composite signal arrive at different times because each frequency propagates at a different speed.
- 1 pt** Noise: unwanted signals that corrupt the signal at the receiving end — thermal (random motion of electrons), induced (motors and appliances), crosstalk (adjacent wires), impulse (spikes from power lines, lightning).
- 1 pt** Decibel: $\text{dB} = 10 \log_{10}(P_2/P_1)$, negative for attenuation and positive for amplification; e.g. half the power is -3 dB .
- 1 pt** SNR = average signal power / average noise power; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$; high SNR means a less corrupted signal (e.g. 10 mW over 1 $\mu\text{W} \rightarrow 10,000 \rightarrow 40 \text{ dB}$).

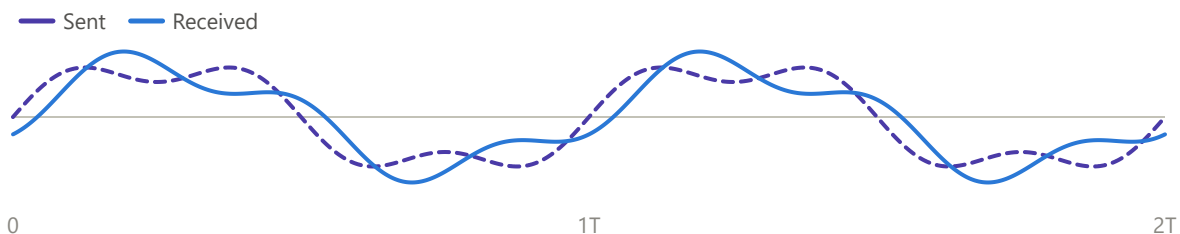
MODEL ANSWER

Attenuation is a loss of energy: a signal, simple or composite, loses some of its energy in overcoming the resistance of the medium. Amplifiers are used to compensate. **Distortion** means the signal changes its form or shape. In a composite signal each frequency component propagates at its own speed, so the components arrive at different times and the received shape no longer matches the one sent. **Noise** corrupts the signal at the receiving end. *Thermal* noise is the random motion of electrons in a wire; *induced* noise comes from motors and appliances that act as sending antennas while the medium acts as a receiving antenna; *crosstalk* is signals jumping from one wire to another, normally adjacent; *impulse* noise is a spike from power lines, lightning and the like.

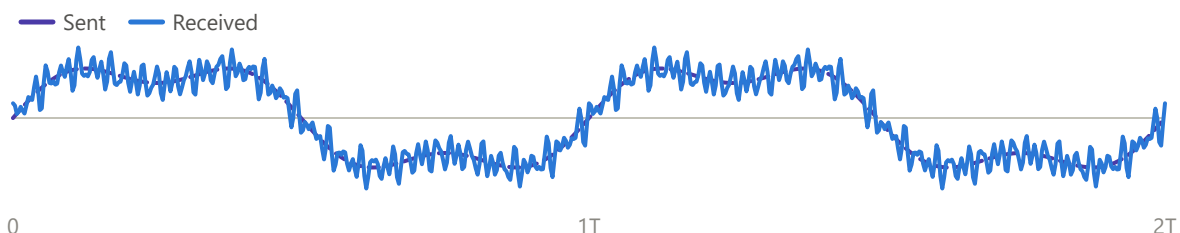
Attenuation — the same shape with less energy (slide 20)



Distortion — the components arrive at different times, so the shape changes (slide 23)



Noise — unwanted energy added on the way (slide 24)



The **decibel** measures the relative strength of two signals or of one signal at two points: $\text{dB} = 10 \log_{10}(P_2/P_1)$. It is negative when the signal is attenuated and positive when it is amplified; halving the power gives $10 \log_{10}0.5 = -3 \text{ dB}$. The **SNR** is the ratio of what is wanted to what is not wanted: $\text{SNR} = \text{average signal power} / \text{average noise power}$, and $\text{SNR}_{\text{dB}} = 10 \log_{10}\text{SNR}$. A high SNR means the signal is less corrupted by noise. Example: 10 mW of signal over 1 μW of noise gives $\text{SNR} = 10,000$ and $\text{SNR}_{\text{dB}} = 40 \text{ dB}$.

L02 pp19-26

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